

A Biologically Inspired, 3D-Printed Multi-DOF Lower-Limb Exoskeleton with High Torque Density Cycloidal Actuation

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Abstract—The goal of this work is to develop a low-cost, biologically inspired lower-limb exoskeleton that better accommodates natural human joint motion. The proposed system provides ten powered and four passive degrees of freedom, enabling multi-planar motion across the hip, knee, ankle, and foot. A modular 3D-printed structure is combined with lightweight, high-torque-density cycloidal actuators to achieve a compact and highly articulated wearable platform. The passive degrees of freedom reduce kinematic constraints and improve anatomical compatibility without requiring additional actuators. Dynamometer testing further demonstrates actuator output torques of 22–23 Nm and a peak torque of approximately 26 Nm, supporting the feasibility of the proposed hardware platform.

Index Terms—biologically-inspired, lower-limb exoskeleton, torque density, 3D-printed.

I. INTRODUCTION

Lower-limb exoskeletons require a careful balance between structural weight, actuator torque capability, and sufficient degrees of freedom to accommodate natural human motion [1], [2]. Many existing rigid systems achieve high torque at the cost of increased mass and mechanical complexity [3], while lightweight designs often sacrifice actuation capability or joint stability. These trade-offs can limit both system performance and anatomical compatibility [3], [4].

To address these challenges, this work presents a low-cost, modular lower-limb exoskeleton that combines a lightweight 3D-printed structure with high-torque-density cycloidal actuators and a multi-DOF architecture. The system incorporates ten powered and four passive degrees of freedom across the hip, knee, and ankle, enabling multi-planar lower-limb motion while maintaining a compact mechanical design. The 3D-printed structure reduces system mass and manufacturing complexity, while the cycloidal actuator design provides high torque density for powered assistance. Together, these features enable a lightweight, high-torque, and highly articulated exoskeleton platform, which is experimentally evaluated using impedance control during squatting.

II. METHODS

The exoskeleton provides fourteen total degrees of freedom (DOFs), including ten powered and four passive DOFs distributed bilaterally across the hip, knee, ankle, and foot. Each leg incorporates five powered DOFs and two passive DOFs. The passive DOFs consist of hip rotation in the transverse

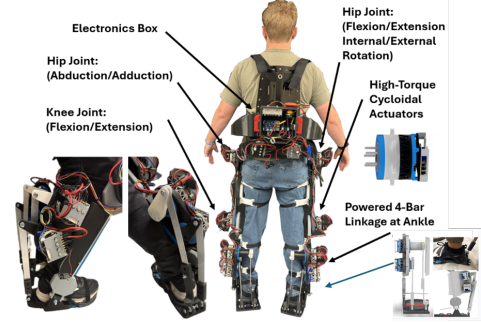


Fig. 1. Annotated rear view of the lower-limb exoskeleton highlighting key structural and actuation components.

plane and foot flexion, which reduce kinematic constraints and allow the exoskeleton to better accommodate natural lower-limb motion without increasing actuator count or system complexity as shown in Fig. 1. At the hip, powered actuation is provided for flexion–extension and abduction–adduction, while a passive hinge permits transverse-plane hip rotation. The knee provides powered flexion–extension. The ankle module uses two motors coupled through dual four-bar linkage mechanisms to generate dorsiflexion–plantarflexion and inversion–eversion. Heim joints and passive foot flexion accommodate coupled motion and reduce mechanical constraints on natural ankle and foot movement.

Most structural components of the exoskeleton were fabricated using additive manufacturing to reduce system mass, manufacturing cost, and fabrication complexity. Load-bearing components were manufactured using PETG or aluminum to maintain structural rigidity, while components interfacing directly with the human body used flexible TPU or commercial harness materials to improve comfort and compliance. Fig. 2 shows the multi-view demonstration of the exoskeleton in different activities.

To provide high joint torque while maintaining a lightweight and compact design, each powered joint uses a modified rolling-pin cycloidal gearbox actuator based on the Open-Mutt platform [5]. The actuator in Fig 3 employs a 13:1 reduction and replaces the sliding contact of conventional fixed-pin cycloidal transmissions with rolling contact, reducing frictional losses and heat generation while retaining the high torque density, distributed load capacity, and shock tolerance of cycloidal transmissions. The original design was further modified for additive manufacturing by using polycarbonate for components in direct thermal contact with the motor to

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Fig. 2. Multi-view demonstration of the exoskeleton in representative postures: (a) standing upright; (b) side view of system layout; (c) squatting; (d) seated; (e) kneeling; (f) ankle four-bar linkage providing active assistance and range of motion with a compliant TPU forefoot joint enabling natural foot flexion.

prevent heat-induced deformation during sustained high-torque operation. Dimensional tolerances between the cycloidal discs and roller elements were also controlled to minimize backlash and improve transmission repeatability. Table I quantitatively compares the proposed actuator with the ULIX [6] and AK90 actuators. At a mass of 0.694 kg, the proposed actuator achieves a nominal torque density of 20.36 Nm/kg, approximately 33.1% and 8.6% higher than ULIX and AK90, respectively. The actuator also achieves a peak torque density of 37.67 Nm/kg and a bandwidth of approximately 60 Hz.

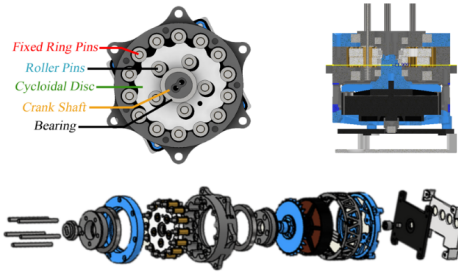


Fig. 3. Compact 3D-printed cycloidal actuator developed for the actuation of the exoskeleton's powered joints, achieving high torque density through a dual-phase cycloidal reduction with rolling pins.

TABLE I
COMPARISON OF ACTUATOR TORQUE DENSITY AND PERFORMANCE

	ULIX	AK90	Cycloidal Drive
Nominal torque density (Nm/kg)	15.3	18.75	20.36
Peak torque density (Nm/kg)	38.8	50.0	37.67
Mass (kg)	0.90	0.960	0.694
Bandwidth (Hz)	3.7	N/A	~60

III. RESULTS AND CONCLUSIONS

To evaluate the functional capability of the proposed lower-limb exoskeleton, dynamic squatting experiments were conducted using an impedance controller. As shown in Fig. 4, the bilateral hip and knee actuators generated approximately symmetric torque profiles, with peak sagittal-plane torques of around 6 Nm at the hip and 5 Nm at the knee. Actuator performance was further characterized using dynamometer testing,

as shown in Fig. 5. The measured output torque approximately followed the theoretical 13:1 transmission relationship, reaching 22–23 Nm at an input torque of approximately 1.7 Nm. Separate maximum-torque testing demonstrated a peak actuator torque of approximately 26 Nm. These preliminary results validate the basic functionality of the proposed exoskeleton and actuator system. Future work will focus on further testing and optimization of the control strategy across a broader range of dynamic movements.

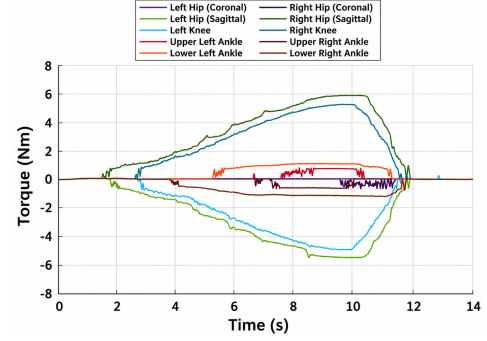


Fig. 4. Actuator torque profiles across the ten powered DOFs during a full-range squat under impedance control. Bilateral hip and knee joints exhibit approximately symmetric torque profiles, with peak sagittal-plane assistance of approximately 6 Nm at the hip and 5 Nm at the knee.

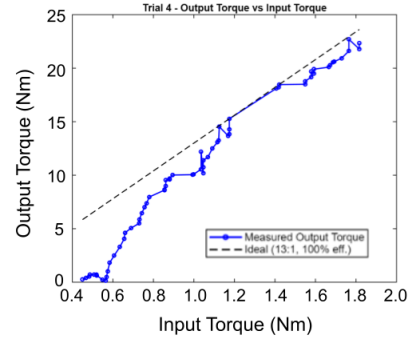


Fig. 5. Experimental characterization of the 3D-printed cycloidal actuator: measured output torque compared with the theoretical 13:1 transmission relationship.

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